



Analog IC Design ASSESSMENT 3

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Registration Number :

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1 Aim

To design and simulate a simple two-transistor NMOS current mirror and a four-transistor cascode current mirror using Cadence Virtuoso, and to study their I_{OUT} versus V_{OUT} characteristics, operating regions, small-signal analysis, and output resistance R_{OUT} .

2 Theory

A current mirror copies a reference current into another branch. The basic idea is to use matched MOSFETs with the same V_{GS} . For identical devices,

$$I_{OUT} \approx I_{REF}. \quad (1)$$

2.1 Simple Current Mirror

The simple mirror uses two NMOS devices. The reference transistor is diode-connected (gate connected to drain). The voltage developed across it fixes V_{GS} , and the same gate voltage is applied to the output transistor. When the output transistor is in saturation, the current is approximately constant.

In a practical MOSFET, channel-length modulation makes the current increase slightly with V_{DS} , so the output current is not perfectly constant.

2.2 Cascode Current Mirror

The cascode mirror adds one more transistor in each branch. The lower pair performs the mirroring, while the upper pair reduces the variation of the lower devices' V_{DS} . The result is a much larger output resistance, but a higher minimum output voltage is required to keep the stacked devices in saturation.

The output resistance used in this experiment is

$$R_{OUT} = \frac{1}{\frac{dI_{OUT}}{dV_{OUT}}} \quad (2)$$

A flatter I_{OUT} – V_{OUT} curve therefore corresponds to a larger R_{OUT} .

3 Simulation Parameters

The values visible in the submitted Cadence schematic are used in this report.

Table 1: Simulation parameters

Parameter	Value
MOS device	N_18_MM
Channel length, L	500 nm
Channel width, W	10 μm
Finger number, N_f	1
Multiplier, m	1
Reference current, I_{REF}	100 μA
Supply voltage, V_{DD}	1.8 V
V_{OUT} sweep for I - V plot	0 to 1.8 V

4 Procedure

4.1 A) To open Virtuoso

1. Open MobaXterm.
2. Click **Session** and open the session settings window.
3. Enter the laboratory credentials and click **OK**.
4. In the terminal, type `ssh` and press Enter.
5. Type `virtuoso &` and press Enter. The Virtuoso log window opens.

4.2 B) To create the library

1. In the Virtuoso log window, select **File** $\rightarrow$ **New** $\rightarrow$ **Library**.
2. Give a library name such as `current_mirror`.
3. Select **Attach to an existing technology library**.
4. Choose **UMC_18_CMOS** and click **OK**.

4.3 C) To create the cell view

1. Open **Tools** $\rightarrow$ **Library Manager**.
2. Select the library created above.
3. Choose **File** $\rightarrow$ **New** $\rightarrow$ **Cell View**.
4. Enter the cell name, for example `current_mirror`.
5. Select **schematic** as the view and click **OK**.

5 Part A: Simple Current Mirror

5.1 1. Circuit Implementation

In the schematic window, place the following components using **Create** → **Instance**.

Table 2: Components for the simple current mirror

Library	Cell	Setting
UMC_18_CMOS	N_18_MM	$W = 10\ \mu m, L = 500\ nm$
analogLib	idc	$I_{DC} = 100\ \mu A$
analogLib	vdc	$V_{DC} = 1.8\ V$
analogLib	gnd	ground

Place two NMOS transistors. Make the left transistor the reference device and connect its gate to its drain. Connect both gates together and connect both sources to ground. Connect the $100\text{-}\mu A$ current source to the reference branch. Connect a DC voltage source to the output drain; this source will later be swept to obtain I_{OUT} versus V_{OUT} .

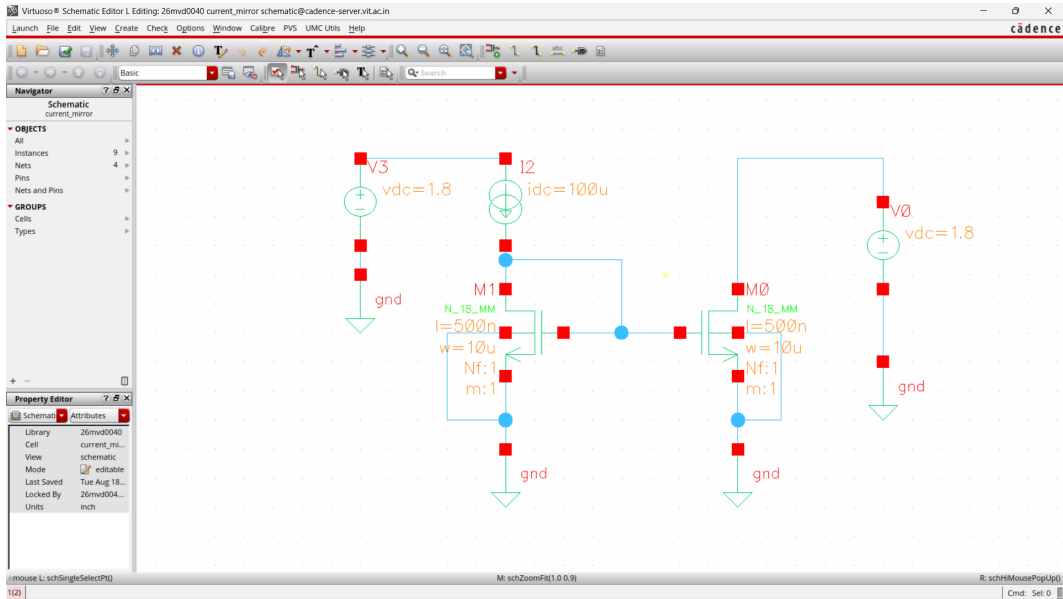


Figure 1: Simple current mirror schematic used for the simulation.

After wiring, click **Check and Save**. If errors are reported, correct the wiring or parameter values before continuing.

5.2 3. Operating Regions from the I_{OUT} - V_{OUT} Characteristic

The operating regions are identified from the knee point of the simulated I_{OUT} - V_{OUT} characteristic. The approximate values read from the graph are summarized below.

Table 3: Operating regions of the simple current mirror from the simulated graph

Operating Region	Approximate V_{OUT} range from the graph
Triode	$V_{OUT} \approx 0$ to $0.2\text{--}0.3\text{ V}$
Saturation	$V_{OUT} \approx 0.2\text{--}0.3\text{ V}$ to 1.8 V
Cutoff	Not observed in the $0\text{--}1.8\text{ V}$ sweep

From the plotted characteristic, the output current rises steeply at low V_{OUT} while the output transistor is in the triode region. Around $V_{OUT} \approx 0.2\text{--}0.3\text{ V}$, the curve reaches its knee and enters the saturation region, where I_{OUT} becomes comparatively less sensitive to further increases in V_{OUT} .

The cutoff region is not observed in the submitted sweep because the output current does not fall to approximately zero over the applied V_{OUT} range.

5.3 4. Open ADE-L

From the schematic window, select **Launch** → **ADE-L**. The Analog Design Environment window opens.

5.4 5. Set the DC Sweep

1. In ADE-L select **Analyses** → **Choose**.
2. Select **dc** and enable the analysis.
3. Select the output source (in the submitted circuit this is $/V0$) as the swept component parameter.
4. Choose parameter **dc**.
5. Set **Start** = **0 V** and **Stop** = **1.8 V**.
6. Click **OK**.

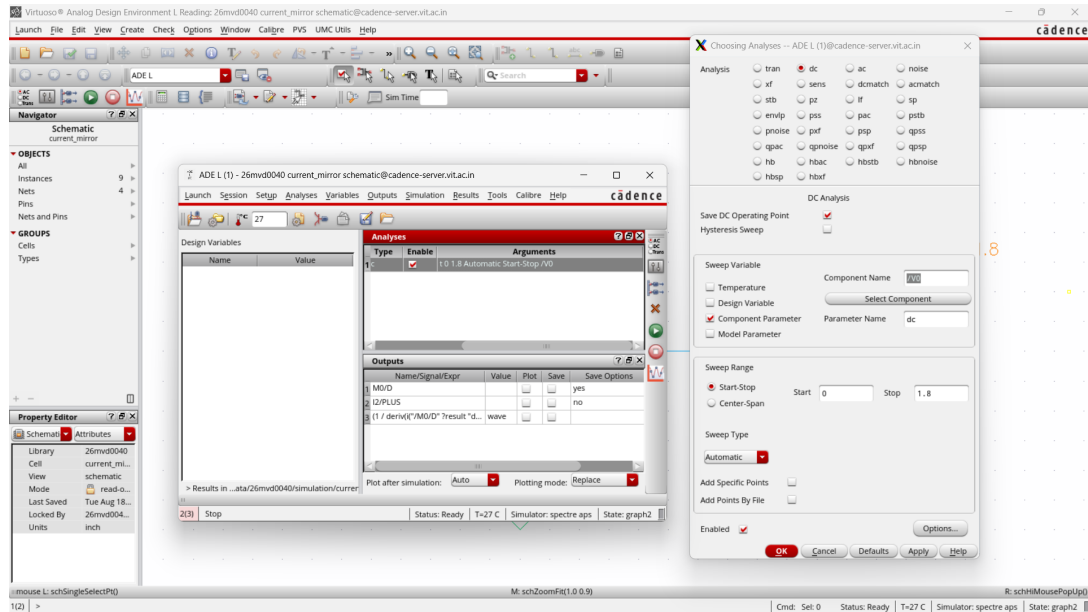


Figure 2: DC sweep settings used in ADE-L.

5.5 6. Select Outputs and Run

1. Select **Outputs** → **To Be Plotted** → **Select on Design**.
2. Click the drain of the output transistor.
3. Also select the reference drain current if a direct comparison with I_{REF} is required.
4. Select **Simulation** → **Netlist and Run**.

5.6 7. I_{OUT} versus V_{OUT} Result

The obtained curve starts with a steep rise at low V_{OUT} and becomes flatter as the output transistor enters saturation.

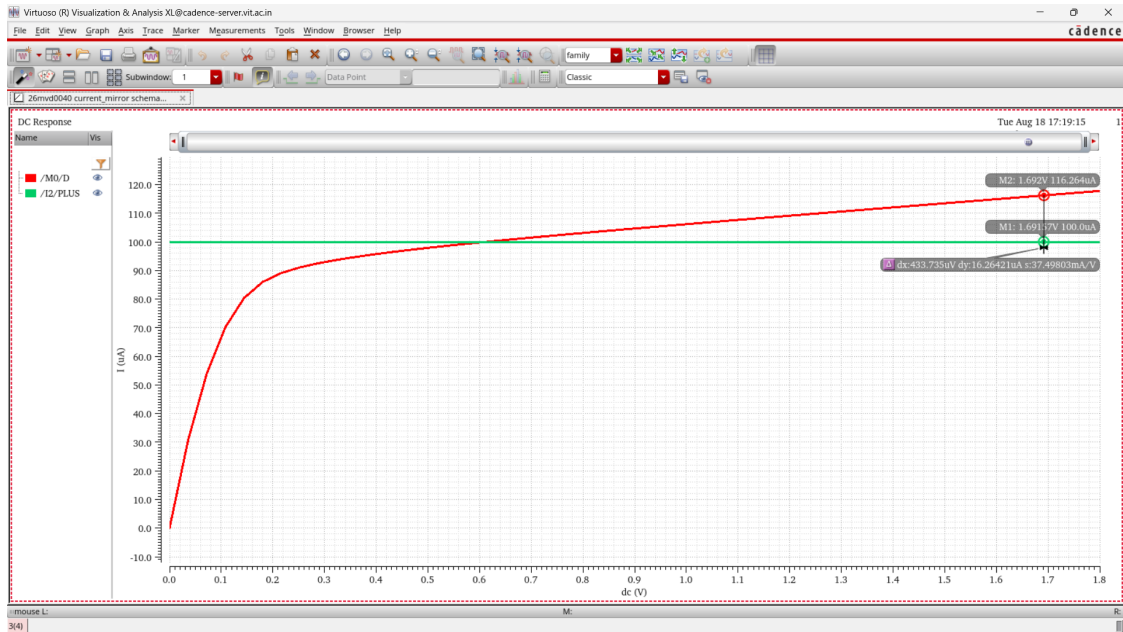


Figure 3: I_{OUT} versus V_{OUT} characteristic of the simple current mirror.

5.7 8. Calculate R_{OUT} in the Calculator

The output resistance is obtained from the slope of the output characteristic.

In the Cadence Calculator use:

```
1/deriv(i("/M0/D" ?result "dc"))
```

Then send the expression to the Outputs section and plot it against the DC sweep variable.

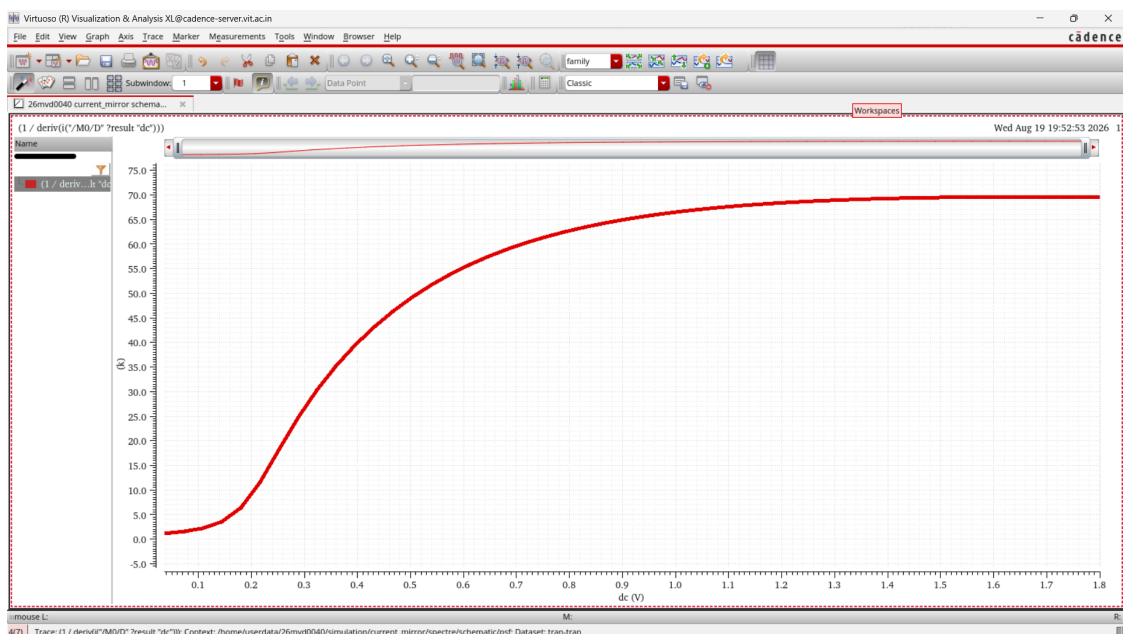


Figure 4: R_{OUT} versus V_{OUT} for the simple current mirror.

From the plotted curve, at $V_{OUT} = 1.8\text{ V}$,

$$R_{OUT, simple} \approx 69.5\text{ k}\Omega. \quad (3)$$

The value is read from the actual plotted curve at the right-hand end of the submitted graph.

6 Part B: Cascode Current Mirror

6.1 1. Circuit Implementation

The cascode mirror uses four identical NMOS devices. In the submitted schematic, the left branch forms the reference stack and the right branch forms the output stack. The lower pair performs the current mirroring and the upper pair provides the cascode action.

Use the same transistor parameters:

$$W = 10\,\mu\text{m}, \quad L = 500\,\text{nm}, \quad N_f = 1, \quad m = 1.$$

Use the same 100- μA reference source and 1.8-V supply.

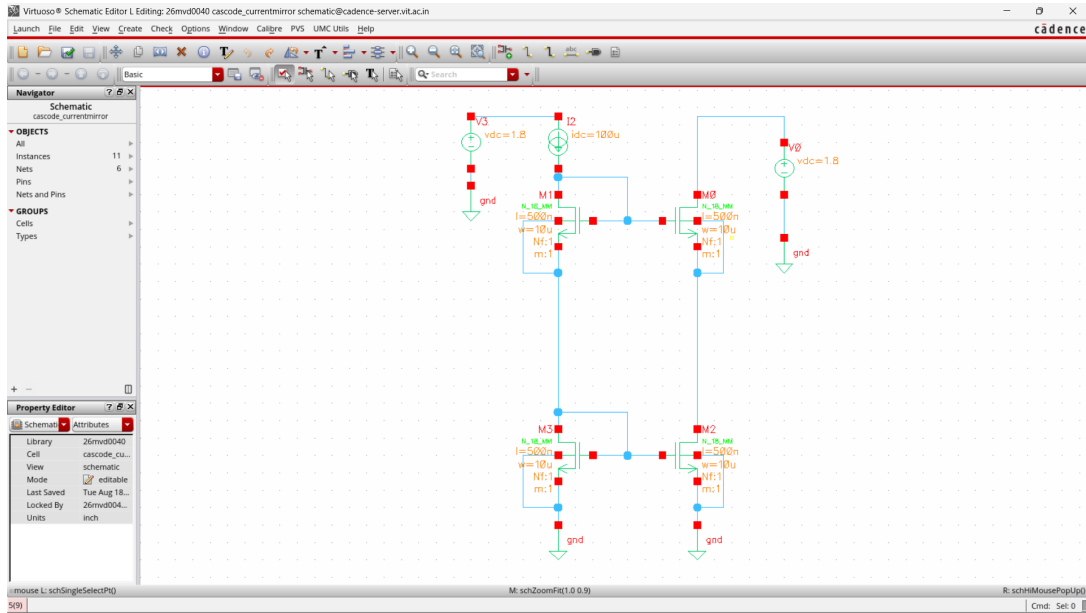


Figure 5: Four-transistor cascode current mirror schematic.

6.2 3. Operating Regions from the I_{OUT} - V_{OUT} Characteristic

The operating regions of the cascode current mirror are identified from the knee of the simulated I_{OUT} - V_{OUT} characteristic. Because the output branch contains stacked devices, the saturation region begins at a significantly larger output voltage than in the simple mirror.

Table 4: Operating regions of the cascode current mirror from the simulated graph

Operating Region	Approximate V_{OUT} range from the graph
Triode	$V_{OUT} \approx 0$ to $1.0\text{--}1.2\text{ V}$
Saturation	$V_{OUT} \approx 1.0\text{--}1.2\text{ V}$ to 1.8 V
Cutoff	Not observed in the $0\text{--}1.8\text{ V}$ sweep

The cascode output current changes significantly at low V_{OUT} because the stacked devices do not initially have enough voltage headroom for all devices to remain in saturation. The curve approaches its saturation-region behavior only after approximately $1.0\text{--}1.2\text{ V}$. The cutoff region is not observed in the submitted sweep because the output current remains non-zero throughout the applied V_{OUT} range.

6.3 4. Connect the Circuit

1. Place four N_18_MM transistors.
2. Form two vertical stacks, one reference stack and one output stack.
3. Connect the lower devices so that the reference current establishes the gate bias for the mirror pair.
4. Connect the upper devices as cascode devices.
5. Connect all required sources to ground and keep the output node at the drain of the upper output transistor.
6. Run **Check and Save** before opening ADE-L.

6.4 5. Configure ADE-L

The DC analysis is the same as for the simple mirror.

1. Launch **ADE-L**.
2. Choose **Analyses** $\rightarrow$ **Choose** $\rightarrow$ **dc**.
3. Sweep /V0 from 0 to 1.8 V .
4. Select the reference and output drain currents in the Outputs section.
5. Run the simulation.

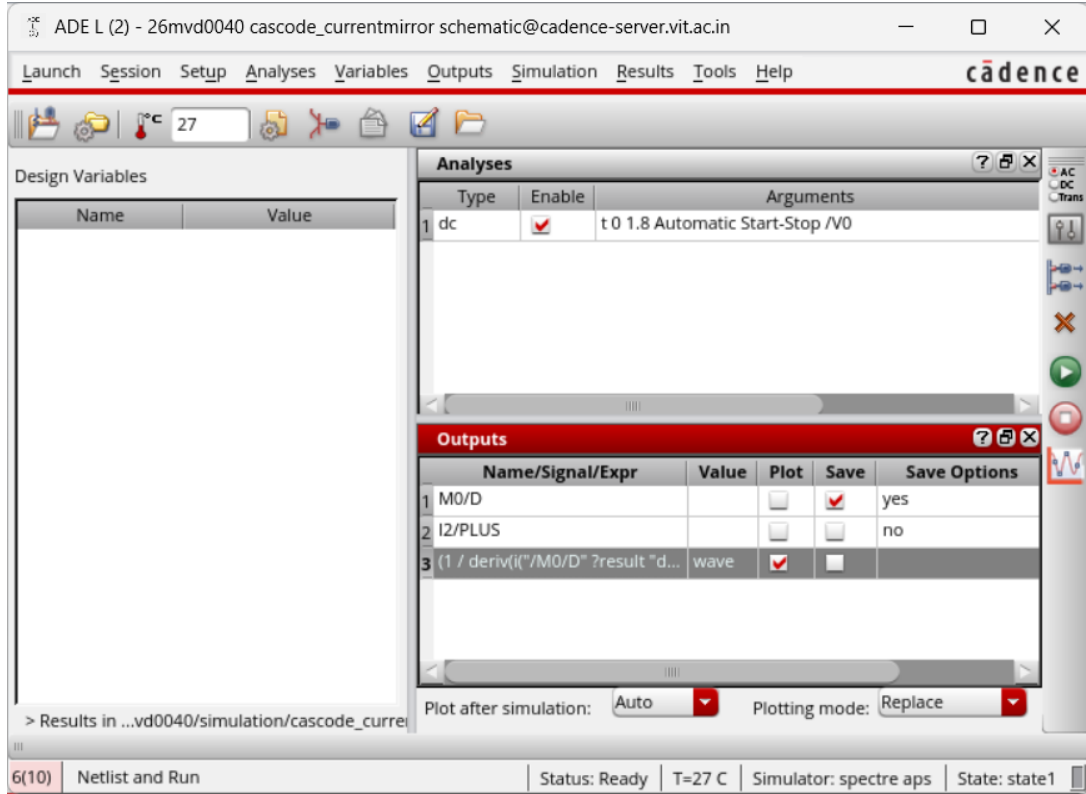
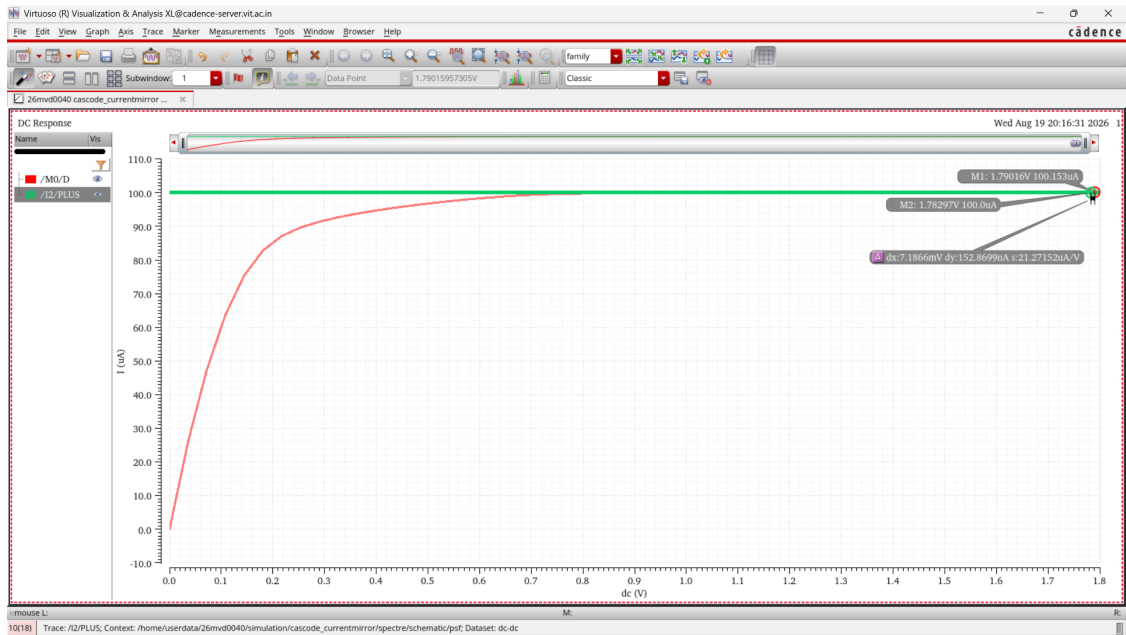


Figure 6: ADE-L configuration for the cascode mirror.

6.5 6. I_{OUT} versus V_{OUT} Result

The output current rises more slowly and approaches the reference current only after a larger output voltage because two stacked output devices must be biased correctly.

Figure 7: I_{OUT} versus V_{OUT} characteristic of the cascode current mirror.

6.6 7. Calculate R_{OUT}

Use the same derivative method:

```
1/deriv(i("/M0/D" ?result "dc"))
```

Plot the expression and read the value at the required output voltage.

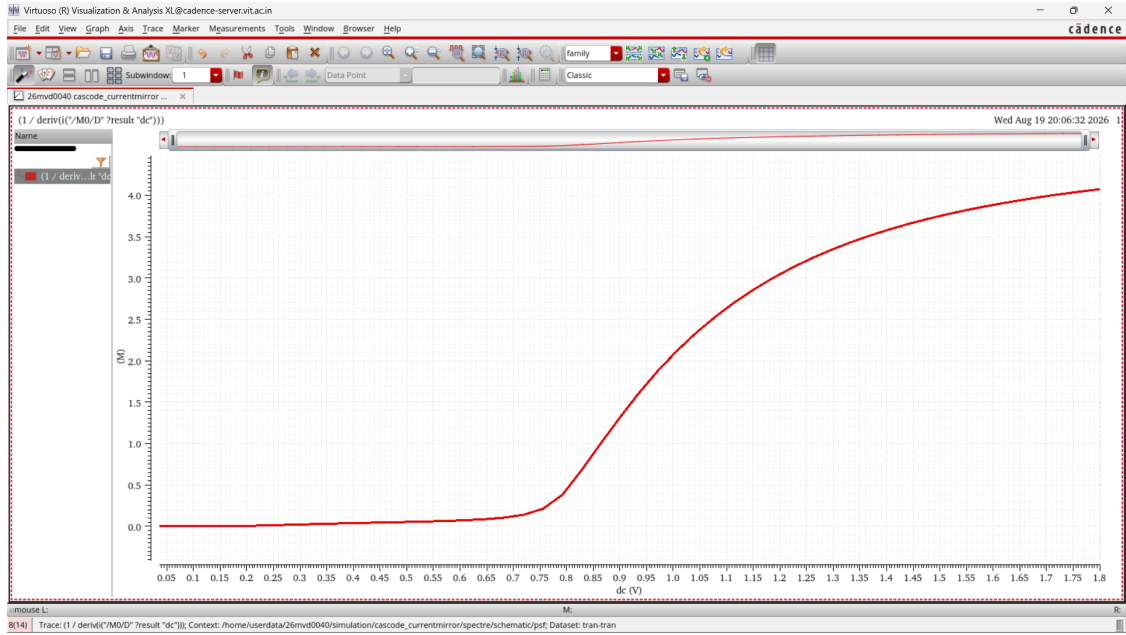


Figure 8: R_{OUT} versus V_{OUT} for the cascode mirror.

From the submitted plot, at $V_{OUT} = 1.8\text{ V}$,

$$R_{OUT, cascode} \approx 4.1\text{ M}\Omega. \quad (4)$$

Therefore, over the same 1.8-V output range, the cascode mirror provides approximately

$$\frac{R_{OUT, cascode}}{R_{OUT, simple}} \approx \frac{4.1 \times 10^6}{69.5 \times 10^3} \approx 59. \quad (5)$$

Hence, the simulated cascode output resistance is about 59 times larger at $V_{OUT} = 1.8\text{ V}$.

7 Results and Observations

Table 5: Measured results from the submitted Cadence plots

Parameter	Simple Mirror	Cascode Mirror
Reference current	$100\ \mu A$	$100\ \mu A$
Supply voltage	$1.8\ V$	$1.8\ V$
W/L	$10\ \mu m/500\ nm$	$10\ \mu m/500\ nm$
R_{OUT} at $V_{OUT} = 1.8\ V$	$\approx 69.5\ k\Omega$	$\approx 4.1\ M\Omega$
Relative R_{OUT}	$1\times$	$\approx 59\times$

7.1 Graph-Based Operating Region Comparison

Table 6: Comparison of operating regions obtained from the simulated output characteristics

Region	Simple Current Mirror	Cascode Current Mirror
Triode	$V_{OUT} \approx 0$ to $0.2\text{--}0.3\ V$	$V_{OUT} \approx 0$ to $1.0\text{--}1.2\ V$
Saturation	$V_{OUT} \approx 0.2\text{--}0.3$ to $1.8\ V$	$V_{OUT} \approx 1.0\text{--}1.2$ to $1.8\ V$
Cutoff	Not observed in the $0\text{--}1.8\ V$ sweep	Not observed in the $0\text{--}1.8\ V$ sweep

These ranges are approximate values read from the simulated $I_{OUT}\text{--}V_{OUT}$ curves. The simple current mirror reaches its saturation region at a much lower output voltage, whereas the cascode mirror requires considerably more voltage headroom.

7.2 Observations

1. In the simple mirror, I_{OUT} changes rapidly at low V_{OUT} while the output transistor is in the triode region.
2. The simple mirror reaches its saturation region at approximately $V_{OUT} = 0.2\text{--}0.3\ V$ based on the knee of the simulated characteristic.
3. Once the simple mirror enters saturation, I_{OUT} becomes nearly constant, although channel-length modulation causes a small positive slope.
4. In the cascode mirror, the output current changes significantly over a larger initial output-voltage range because the stacked devices require greater voltage headroom.

5. The cascode mirror reaches its saturation region at approximately $V_{OUT} = 1.0\text{--}1.2\text{ V}$ based on the knee of the simulated characteristic.
6. The cascode topology therefore provides a much larger output resistance than the simple mirror.
7. The price paid for the higher R_{OUT} is higher voltage headroom before the output branch reaches its intended saturation-region operation.
8. Cutoff is not observed for either mirror in the submitted $0\text{--}1.8\text{ V}$ sweep because the output current does not reduce to approximately zero.

7.3 Important Note About the R_{OUT} Reading

The values above are taken from the two dedicated R_{OUT} plots generated from the submitted Cadence data. They are therefore more appropriate for this report than using the spacing between two markers on the $I\text{--}V$ plot. The marker spacing on the $I\text{--}V$ plot compares two traces, whereas R_{OUT} requires the local reciprocal slope $1/(dI/dV)$ of the output characteristic.

8 Conclusion

The simple and cascode current mirrors were successfully designed and simulated in Cadence Virtuoso. The simple mirror provides lower voltage headroom but a much smaller output resistance. The cascode mirror requires more headroom but gives approximately $4.1\text{ M}\Omega$ at 1.8 V , compared with $69.5\text{ k}\Omega$ for the simple mirror.

The $I_{OUT}\text{--}V_{OUT}$ characteristics also show the difference in operating regions. The simple mirror enters the saturation region at approximately $0.2\text{--}0.3\text{ V}$, while the cascode mirror requires approximately $1.0\text{--}1.2\text{ V}$ before reaching its saturation-region behavior. The cutoff region is not observed in the applied $0\text{--}1.8\text{ V}$ sweep.

The simulation therefore verifies the fundamental trade-off between voltage headroom and output resistance in current-mirror design.